

Earth's hemispheres gets a little more, or a little less, solar radiation than they're used to.

And that is where the fun begins.

Milanković's theory initially assumed that a tilt of the Earth's hemispheres caused ravenous winters cold enough to turn the planet into ice. But a Russian meteorologist named Wladimir Köppen dug deeper into Milanković's work and discovered a fascinating nuance.

Moderately cool summers, not cold winters, were the icy culprit.

It begins when a summer never gets warm enough to melt the previous winter's snow. The leftover ice base makes it easier for snow to accumulate the following winter, which increases the odds of snow sticking around in the following summer, which attracts even more accumulation the following winter. Perpetual snow reflects more of the sun's rays, which exacerbates cooling, which brings more snowfall, and on and on. Within a few hundred years a seasonal snowpack grows into a continental ice sheet, and you're off to the races.

The same thing happens in reverse. An orbital tilt letting more sunlight in melts more of the winter snowpack, which reflects less light the following years, which increases temperatures, which prevents more snow the next year, and so on. That's the cycle.

The amazing thing here is how big something can grow from a relatively small change in conditions. You start with a thin layer of snow left over from a cool summer that no one would think anything of and then, in a geological blink of an eye, the entire Earth is covered in miles-thick ice. As glaciologist Gwen Schultz put it: "It is not necessarily the amount of snow that causes ice sheets but the fact that snow, however little, lasts."

The big takeaway from ice ages is that you don't need tremendous force to create tremendous results.

If something compounds—if a little growth serves as the fuel for future growth—a small starting base can lead to results so extraordinary they seem to defy logic. It can be so logic-defying that you underestimate what's possible, where growth comes from, and what it can lead to.

And so it is with money.

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